5.4.2. AC circuits with pure Inductance

A circuit having only inductance and without resistance and capacitance is called pure Inductance circuit.

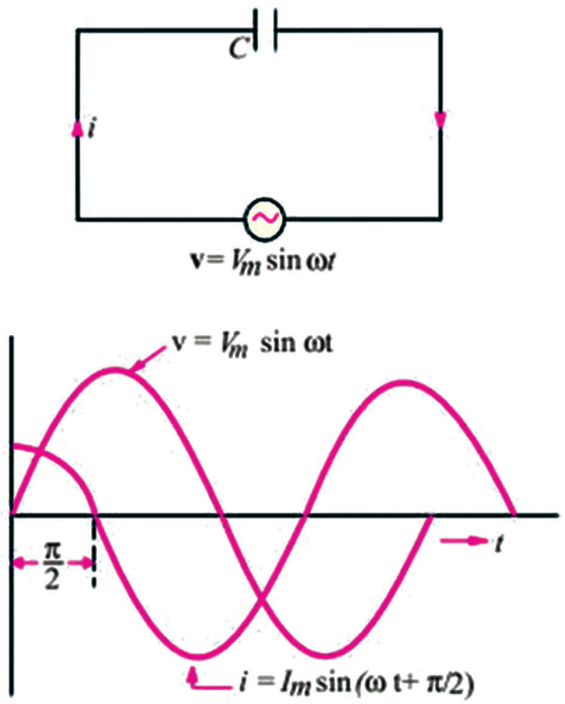


Fig 5.8 *AC through Inductance*

If A.C current flows through a coil, Back emf is induced due to inductance of the coil. This Back emf opposes supplied voltage in a pure inductance coil. Back emf is equal to supply voltage. In inductive circuit only, the frequency is same for voltage and current, but they are out of phase and current is lagging by 90° to the voltage.Therefore powerfactor (cos 90=0)is zero.

5.4.3. Inductive Reactance

Inductive reactance means the opposition due to self inductance to the A.C current through a coil. It's unit is ohm and it is denoted by the letter "XL"

$$XL = 2\pi fL$$

Where

XL-Inductive reactance in ohm
f-frequency in Hertz
L-Inductance in Henry

5.4.4. AC Circuits with pure capacitance

Two conducting plates separated by insulating material is called as capacitor. The insulating materials are be air, mica and paper.

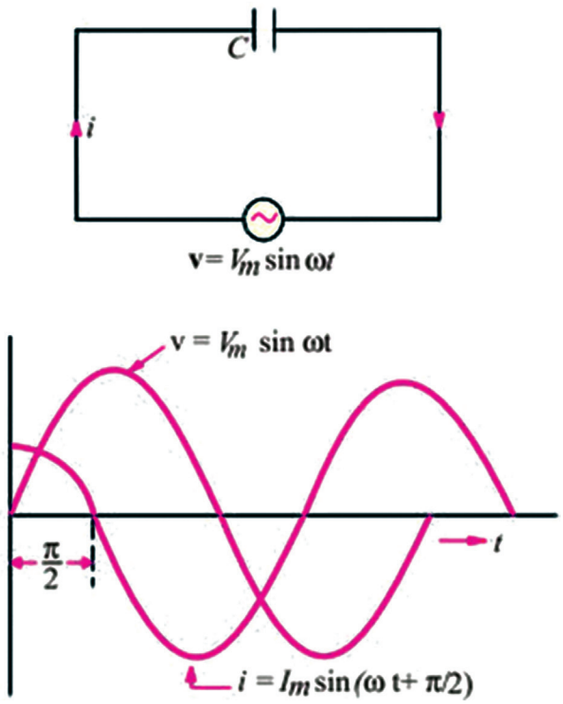


Fig 5.9 *AC through Capacitance*

In the first half cycle (up to 90°) capacitor is charged and from 90° to 180° the capacitor is discharged. Similarly in the second half cycle, capacitor is charged first and discharged next, in opposite direction. So, in one cycle, capacitor is charged and discharged two times, in capacitor only. In A.C circuit, the current is leading by the voltage at 90°

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